

8 Patients with cystic fibrosis may develop serious bacterial infections.

A patient with cystic fibrosis developed bacterial infections including *Mycobacterium*.

(a) (i) This patient was given a combination of antibiotics for several months.

Explain why a combination of antibiotics had to be given for several months.

(2)

(ii) This patient did not respond to the combination of antibiotics, and later needed a lung transplant.

Suggest why this patient needed a lung transplant.

(3)

(iii) Following the transplant, the patient was given immunosuppressive drugs.

Immunosuppressive drugs weaken the immune system. Some of these drugs work by preventing DNA synthesis in the patient.

As a result of the immunosuppressive drug treatment, the infection with *Mycobacterium* developed faster.

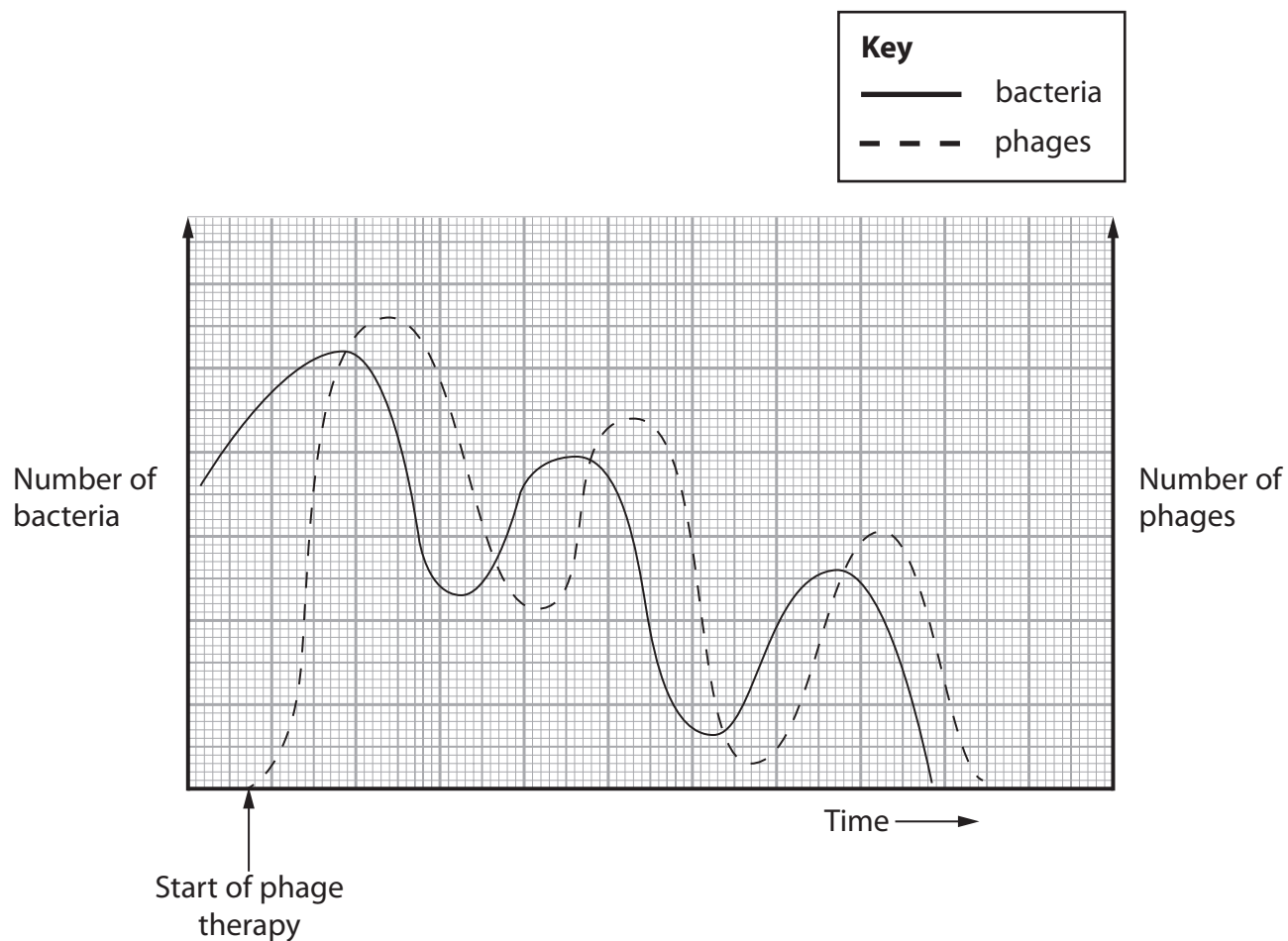
Explain why the infection with *Mycobacterium* developed faster when the patient was taking immunosuppressive drugs.

(4)

*(b) Phage therapy was used to treat the *Mycobacterium* infection in this patient.

Phages are viruses that target bacterial cells.

The graph shows how the number of *Mycobacterium* and the number of phages changed following the start of phage therapy.



Use the information in the graph to support your answer.

[illegible]**TOTAL FOR PAPER = 90 MARKS**